Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

Advanced Diploma in Information & Communication Technology

PROFESSIONAL DEVELOPMENT/METHODS

PFD470S/PFD471S/PFD472S

SUBJECT GUIDE:	2025
COURSE CODE:	ADICTA
NQF LEVEL:	7
PREREQUISITE SUBJECTS:	PREREQUISITE SUBJECTS: Diploma NQF Level 6

Table of Contents

1. INTRODUCTION

Welcome to the Advanced Diploma module for Professional Development/Methods. The purpose of this course is to ensure that you acquire basic professional skills pertaining to time and money management in the context of project management and quality control.

2. GENERAL

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator	T. Makhurane	Engineering 2.42	(021) 460 3178	makhuranet@cput.ac.za
Lecturers	T. Makhurane	Engineering 2.42	(021) 460 3178	makhuranet@cput.ac.za
Domain Coordinator	T. Makhurane	Engineering 2.42	(021) 460 3178	makhuranet@cput.ac.za
Head of Department	T. Ncubukezi	Engineering 2.30	(021) 460-3713	ncubukezit@cput.ac.za
Secretary	N. Allie	Engineering 2.1	(021) 460-3010	alien@cput.ac.za
Support Technician				
Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za
Support Technician	I. Saliem	Engineering 1.28	(021) 460-3031	saliemi@cput.ac.za
Support Technician				

2.2 Class Times and Venues

As published on http://myclassroom.cput.ac.za and notice boards. The subject consists of one online session every other week. Additional classes may be scheduled however, as and when necessary.

2.3 Learner Management System (LMS)

https://myclassroom.cput.ac.za Blackboard used for general communications and announcements, posting of course material, and assignments. It is the responsibility of the learner to check Blackboard regularly.

2.4 Crims System

https://enquiries.cput.ac.za The crims system is an online platform to help students open tickets and raise subject and other academic related queries. Students must not open a crims ticket if the matter can be addressed with the Lecturer/Subject Coordinator directly.

3. STUDY MATERIALS

3.1 Prescribed Textbook

A number of readings will be recommended by the lecturer and included on slides and from time to time. The following may prove useful but are not essential:

- Any academic text on data structures and algorithms
- Any academic text on discrete mathematics

3.2 Recommended Readings

Material on Data Structures and Algorithms is available on many university websites. In addition, many books have been published on Data Structures and Algorithms. The lecturers will be pointing out readings of interest to the students as the subject progresses

3.3 Websites

Any websites on data structures and algorithms

Any websites on discrete mathematics

4. MODULE CREDITS

This module has 20 SAQA or 0.167 HEMIS credits on NQF level 7. Please refer to the Faculty prospectus on http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 200 notional hours allocated to this module. These notional hours comprise of 60 hours (30%) for contact time and the remainder will be self-study and other tasks, assignments, and tests.

METHOD	% OF YOUR LEARNING TIME
Seminars / Practicals (face-to-face, limited interaction or technologically mediated)	30%
Independent self-study of standard texts and references and specially prepared materials (study guides, books, journals, using online tutorials etc.)	20%
Independent self-study, time needed to prepare for assessment (e.g. tests/exams)	25%
Student learning groups (working with your peers for group assignments)	25%

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Kindly refer to CPUT's Policy Docs and Student Guidelines:

https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf

5.2 Class attendance and participation

Attendance is compulsory. Kindly refer to page 19 of CPUT's General Handbook Academic and Student Rules and Regulations: https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf

5.3 Grievance procedures

Kindly refer to page 76 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf

5.4 Plagiarism

Kindly refer to page 29 of CPUT's General Handbook Academic and Student Rules and Regulations: https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf

6. ASSESSMENT

6.1 Assessment policy and regulations

This subject has 3 major tests. An additional major assessment item will be an individual project. Minor assessment items in the form of assignments, quizzes and exercises may also be issued to aid in the development of skills as and when needed.

Tests will take the form of written tests taken in a classroom or examination hall. While an assignment will generally have a one week submission time using Blackboard. Projects will have longer times.

Submissions can include scanned versions of handwritten work, or typed work presented as .docx or .pdf files.

Test queries: Last day for any queries regarding a test is one week after the test was discussed in class. Thereafter the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be given to your lecturer (via the Secretary) within one week after the test was written. You must also inform your lecturer on the day of the test that you are ill.

Due dates must be strictly adhered to. **NO late submissions are allowed for any subject in this programme.**

Assignments may be administered via Blackboard and other platforms.

6.2 Assessment Opportunities: Administration

The Table below summarises the marks weighting.

ASSESSMENT BREAKDOWN AND WEIGHTS					
ASSESSMENT CRITERIA	**1ST TERM**	**2ND TERM**	**3RD TERM**	**4TH TERM**	**TOTALS**
Nature of assessment	Test (T1)	Test (T2)	Test (T3)	Test (T4)	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	
Total assessment weight for the term	10%	35%	20%	35%	100%
Will the assessment be moderated?	No	Yes	No	*Yes	

STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

Topics included in this Capita Selecta include items drawn from Data Structures and Algorithms and equally importantly the Analysis and comparison pertaining to the performance of algorithms. For implementation purposes students can make use of an object-oriented language. We will work primarily with Java. Alternatively, Python or C# may also be used.

7.2 Subject Structure

7.2.1 Subject Outcomes

SUBJECT OUTCOMES	DESCRIPTION
SO1	Introduction to Project Management
SO2	Introduction to Management Henri Fayol Henry Mintzberg
SO3	Planning and Scheduling Work Breakdown Structure Network Analysis using CPM and PERT Gantt charts
SO4	Earned Value Analysis Resource allocation and smoothing Managing the project baseline
SO5	Project Selection Basic Investment Calculations
SO6	Project Organization and Communication Definition Start Steady State Termination
SO7	Software Development Lifecycles Waterfall Spiral V-model Evolutionary models

SO8	Configuration Management Configuration Item identification Promotion Management Release Management Branch Management Variant Management Change Management
SO9	Rationale Management Value of Decision Justification Recording Rationale
S10	Risk Management Definition of Risk How Risk Arises How Risk can be addresse
S11	Software Project Management Plan Raison d'être Overview of the Plan
S12	Software Estimation Purpose and Value Lines of Code techniques (illustrated by COCOMO 81) Introduction to Contemporary Methods Function Point techniques (illustrated by Albrecht's Method) Introduction to Contemporary Methods
S13	Quality ISO Capability Maturity Model Project Audits When, Why and by Whom Testing

S14	Human Capital Topics
S15	Purchasing Concepts
S16	PMBOK Review
S17	Change Management
S18	Governance and Compliance
S19	Professional Practice

STUDY UNIT 3: SERIES, SEQUENCES, MATRICES				
SO5				
	Vectors, Arrays, Matrices: Mathematical viewpoint and Programmatic viewpoint.			
TERM 2				
PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES: •				
SO6				
	Data Structures, Problems, Input Size, Algorithms versus programs, Searching and Sorting Problem, Algorithm variants.			
	Foundational Data Structures: Abstract Data Type, Linked List Doubly l=Linked list, Stack and Queue. Implementation issues for dynamic structures (linked structures) versus static structures (arrays), Recursion.			
SO8				
	Analysis of Algorithms I: Complexity Analysis, Comparing Algorithms, Growth Rate of Functions (Asymptotic notation). Some Properties of Big O notation. Showing f is O(g), Showing f is not O(g).			
SO9	Trees: Tree ADT, Linked Tree Representation, Tree Property, Computing Height of a Tree, Tree Traversals			
SO10	Binary Search Trees: Ordered Dictionary ADT, BST Implementations, Traversal, Smallest, Get, Put, Remove, Successor.			
SO11	AVL Trees: Height, AVL Trees, Re-Balancing AVL Trees, , AVL Tree Performance.			
SO12	Graphs: Degrees and the Handshaking Lemma, Complete Graphs, Paths and Cycles, Trees, Forests, Subgraphs, and Connectivity, Graph Representations. Graph Traversals: Depth-First Search (DFS), Path-Finding, Cycle Detection, Counting Vertices, DFS Tree, Breadth-First Search (BFS), Summary. Minimum Spanning Trees: Weighted Graphs, Minimum Spanning Trees & Algorithms, Kruskal's algorithm, Disjoint set based implementation of Kruskal's algorithm, Prim's Algorithm, Heap-Based Implementation of Prim's Algorithm. Shortest Paths: Single-Source Shortest Path Problem, Dijkstra's Algorithm.			
SO13	Multiway Search Trees: Beyond Binary Search Trees, Get, Put, Successor and Remove, (2,4)-Trees, B-Trees.			
SO14	Hashing: Hash Function, Hash Code, Separate Chaining, Open Addressing, Revisiting the Load Factor.			

SO15	Priority Queues & Heaps: Priority Queues, Heaps, Array-Based Implementation, Building a Heap, Application: Sorting, Introduction to Amortized Analysis			
SO16	Dictionaries: The Dictionary Problem, Simple Implementations of a Dictionary.			
SO17	Huffman encoding			
SO18	Brief Introduction to Automata Theory && Regular Expressions, Theoretical Computer Science concepts.			

7.3 Articulation with other subjects in the programme

This subject is not directly linked to a subject in Post Graduate Diploma.

7.4 Learning presumed to be in place

Diploma or any other equivalent NQF level 6 qualification.

7.5 Critical Cross-field Outcomes

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display responsible decisions, using critical thinking, have been made.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical, and/or language skills in the modes of written and/or oral presentation.
CCFO6	Use science and technology effectively and critically, showing responsibility towards the environment and the health of others.
CCFO7	Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.
CCFO8	Contribute to the full personal development of each learner and the social and economic development of the society at large, by making it the underlying intention of any programme of learning to create awareness in individuals about the importance of: i) reflecting on and exploring a variety of strategies to learn more effectively; ii) participating as responsible citizens in the life of local, national and global communities; and iii) being culturally and aesthetically sensitive across a range of social contexts. (RSASAQA, 2001:24)

8. STUDY UNITS/STUDY THEMES

8.1 Study Units

Refer to 7.2.2 Subject Breakdown Structure.

8.2 Assessment Opportunities and Assessment strategy

Refer to 6.2 Assessment Opportunities: Administration for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.3 Subject Outcomes and Associated Assessment Criteria

Refer to 7.2.2 Subject Breakdown Structure for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.4 Self-study Activities

Exercises and quizzes.

9. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY STUDENT ACKNOWLEDGEMENT

2025

ADVANCED DIPLOMA in INFORMATION and COMMUNICATION TECHNOLOGY: APPLICATIONS DEVELOPMENT (ADICTA)

PROFESSIONAL DEVELOPMENT-METHODS 4 (PFD470S/PFD471S/PFD472S)

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **PROFESSIONAL DEVELOPMENT-METHODS 4** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname: ………………………………………………..………
(Print)

Student Number: …………………………………

Contact number: ……………………… (home/res) Contact number: ……………………… (cell)

Signature: ……………………… Date: ………………………